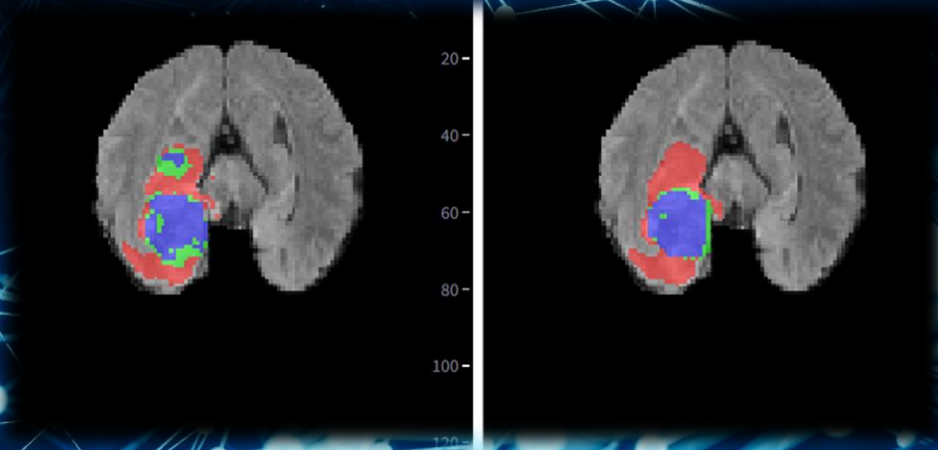
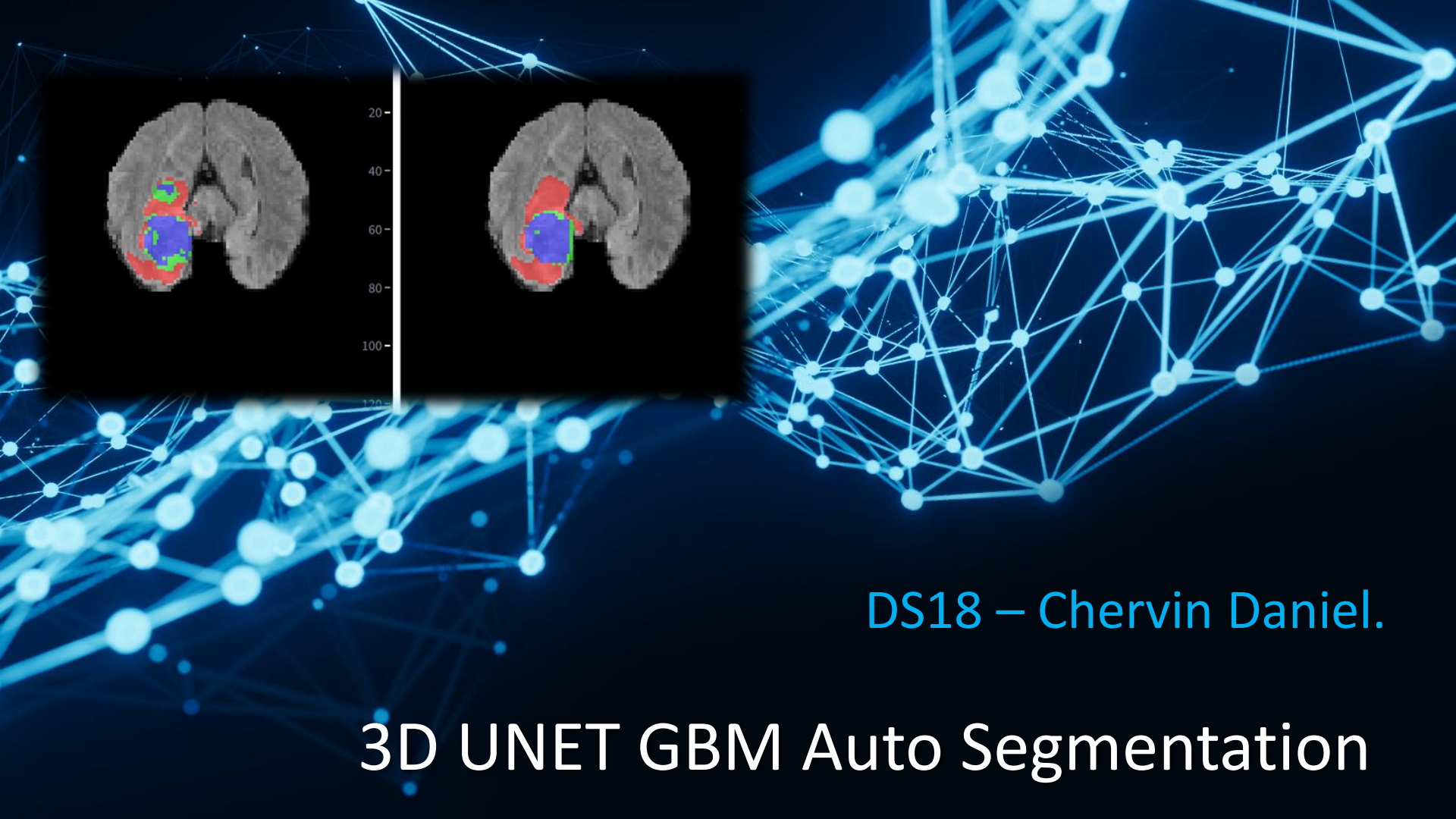
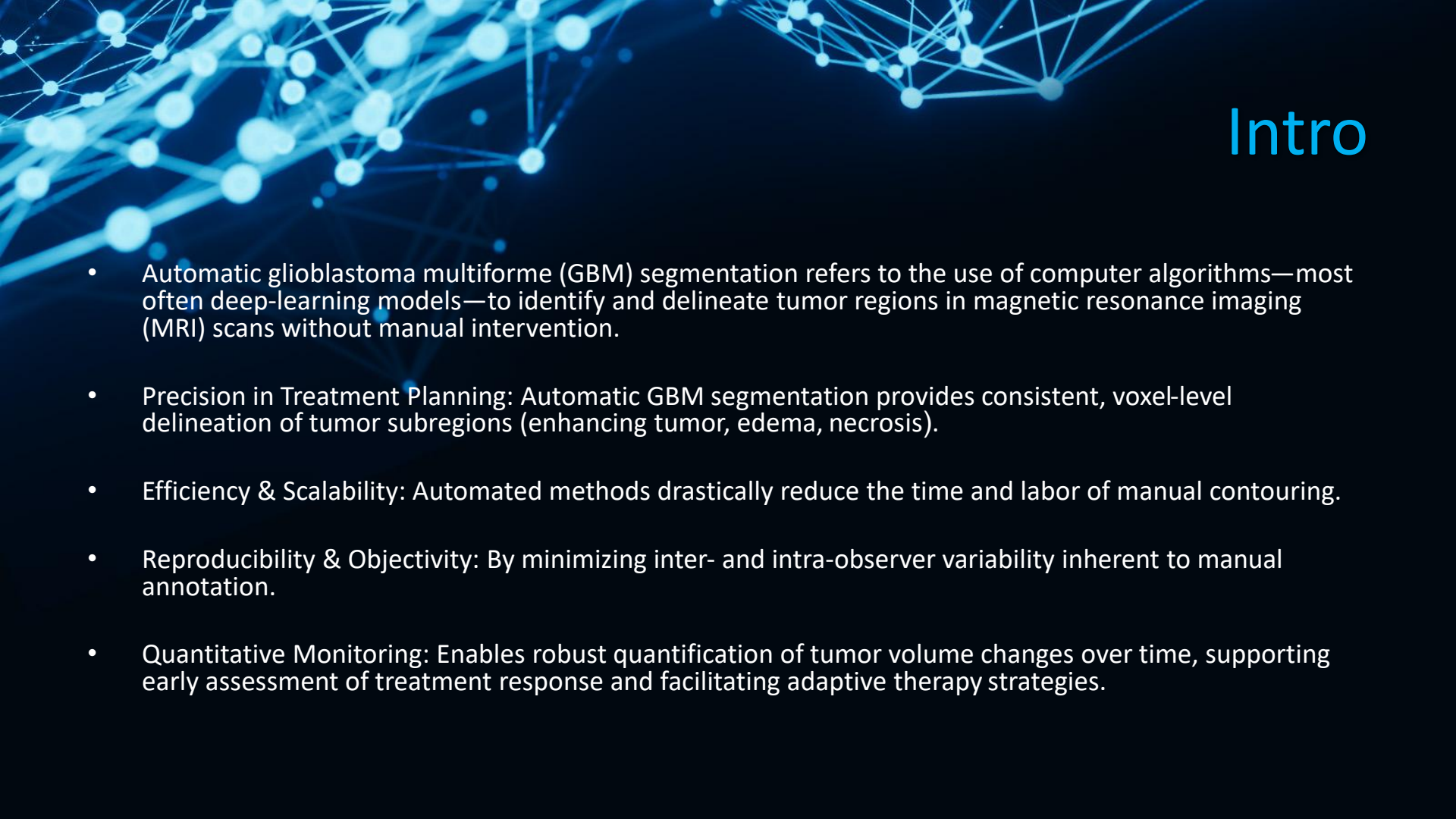




DS18 – Chervin Daniel.

3D UNET GBM Auto Segmentation



Intro

- Automatic glioblastoma multiforme (GBM) segmentation refers to the use of computer algorithms—most often deep-learning models—to identify and delineate tumor regions in magnetic resonance imaging (MRI) scans without manual intervention.
- Precision in Treatment Planning: Automatic GBM segmentation provides consistent, voxel-level delineation of tumor subregions (enhancing tumor, edema, necrosis).
- Efficiency & Scalability: Automated methods drastically reduce the time and labor of manual contouring.
- Reproducibility & Objectivity: By minimizing inter- and intra-observer variability inherent to manual annotation.
- Quantitative Monitoring: Enables robust quantification of tumor volume changes over time, supporting early assessment of treatment response and facilitating adaptive therapy strategies.



Datasets & repositories

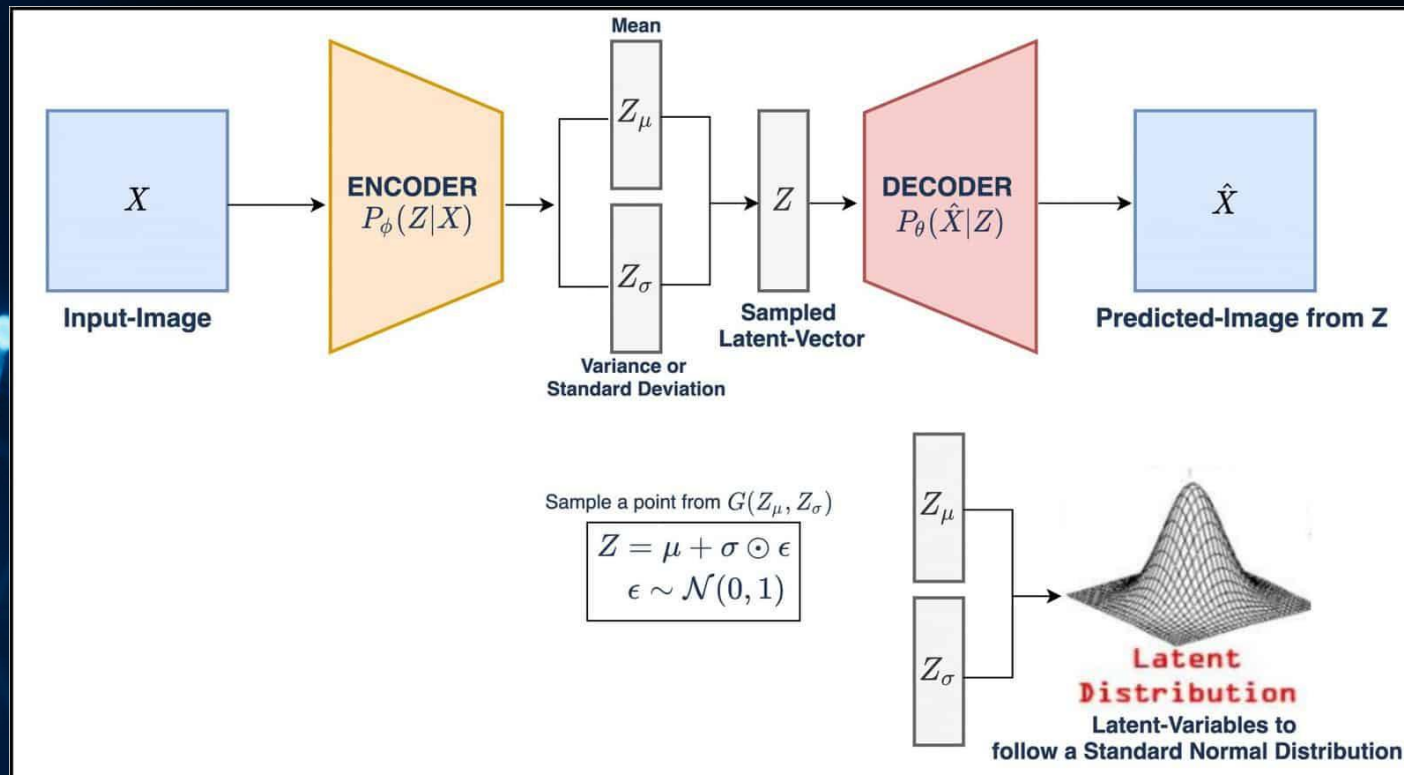
- Healthy Brain full scans in nifty format.
 - IXI dataset <https://brain-development.org/ixi-dataset/>
- Brain full scans with GBM + labels (segmentation) nifty format.
 - BRATS dataset <https://www.cancerimagingarchive.net/collection/ucsd-ptgbm/>
- Github repo
 - https://github.com/daniel-chervin/DS18_FinalProject/tree/main
- Streamlit app
 - <https://ds18finalproject-danielc.streamlit.app/>

An abstract graphic on the left side of the slide, consisting of a dense network of glowing blue nodes connected by thin, light blue lines, resembling a neural network or a complex data structure.

Two Strategies tested

- Unsupervised (private interest)
 - Train Autoencoders on healthy brain MRI scans (IXI).
 - Inference on Brain scan with tumor and analyze reconstruction anomalies.
 - Does not require a labeled dataset, advantage.
 - Spoiler... not accurate enough segmentation, disadvantage.
- Supervised
 - Train a 3D UNET CNN on MRI scans with GBM + labels dataset.
 - Evaluate on a ground truth scans with labels.

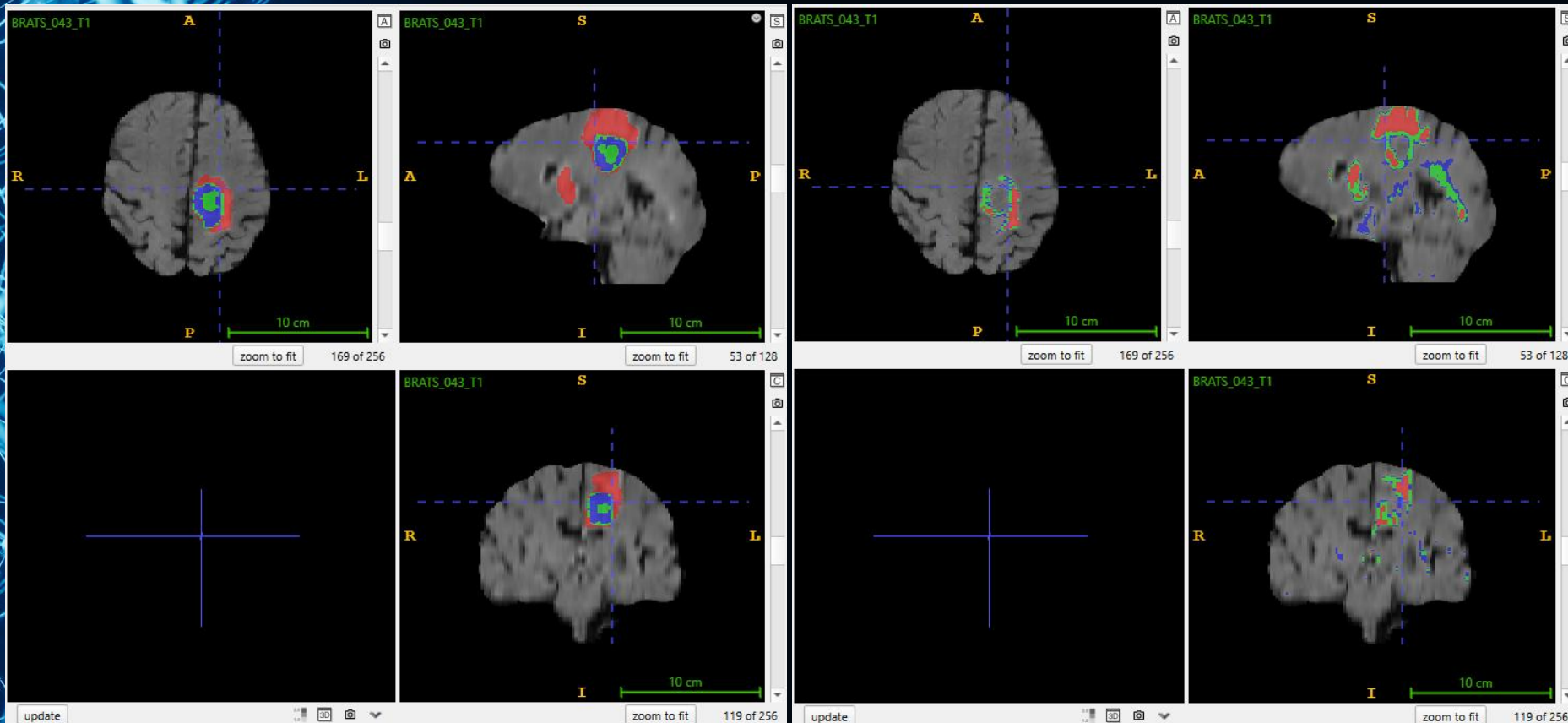
Unsupervised – VAE with MB Conv



Unsupevised – VAE with MB Conv

- **Input volume:** A 3D scan enters the encoder.
- **Encoder path:**
 1. Series of MBConv3D blocks that expand channels and down-sample via stride.
 2. Global average pooling yields a single feature vector.
 3. Two linear layers produce μ and $\log \sigma^2$ for the latent distribution.
- **Latent space (z):** Sampling occurs via the reparameterization trick.
- **Decoder path:**
 1. Project z back to a small feature map.
 2. Three rounds of ConvTranspose3D up-sampling followed by MBConv3D refinement.
 3. Final $1 \times 1 \times 1$ convolution reconstructs the single-channel volume.

VAE MBConv results



An abstract graphic on the left side of the slide, featuring a complex network of glowing blue nodes and connecting lines, resembling a neural network or a molecular structure, set against a dark blue background.

VAE MBConv conclusions

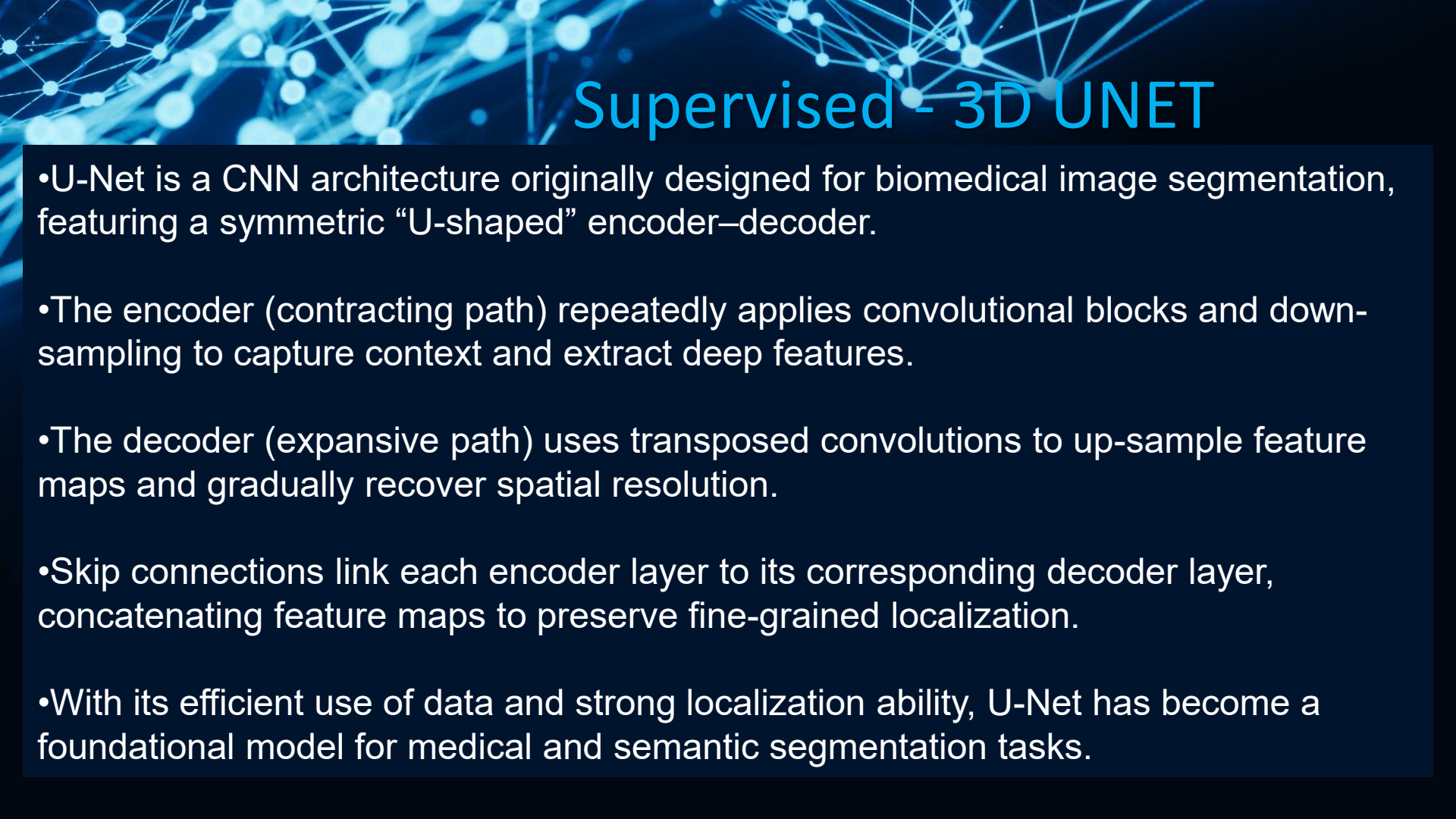
- VAE + MBConv model is consistently picking up the high-contrast core of the tumor
- Under-segmentation, misses central and peripheral regions.
- Spurious false positives, small, scattered green/blue islands in regions where the ground truth has none.
- Shape irregularity, jagged output, holes...

Next steps (post DS18 ;)

- Increase latent capacity.
- Morphological post-processing, eliminate spurious islands and fill holes.
- Etc...

Overall

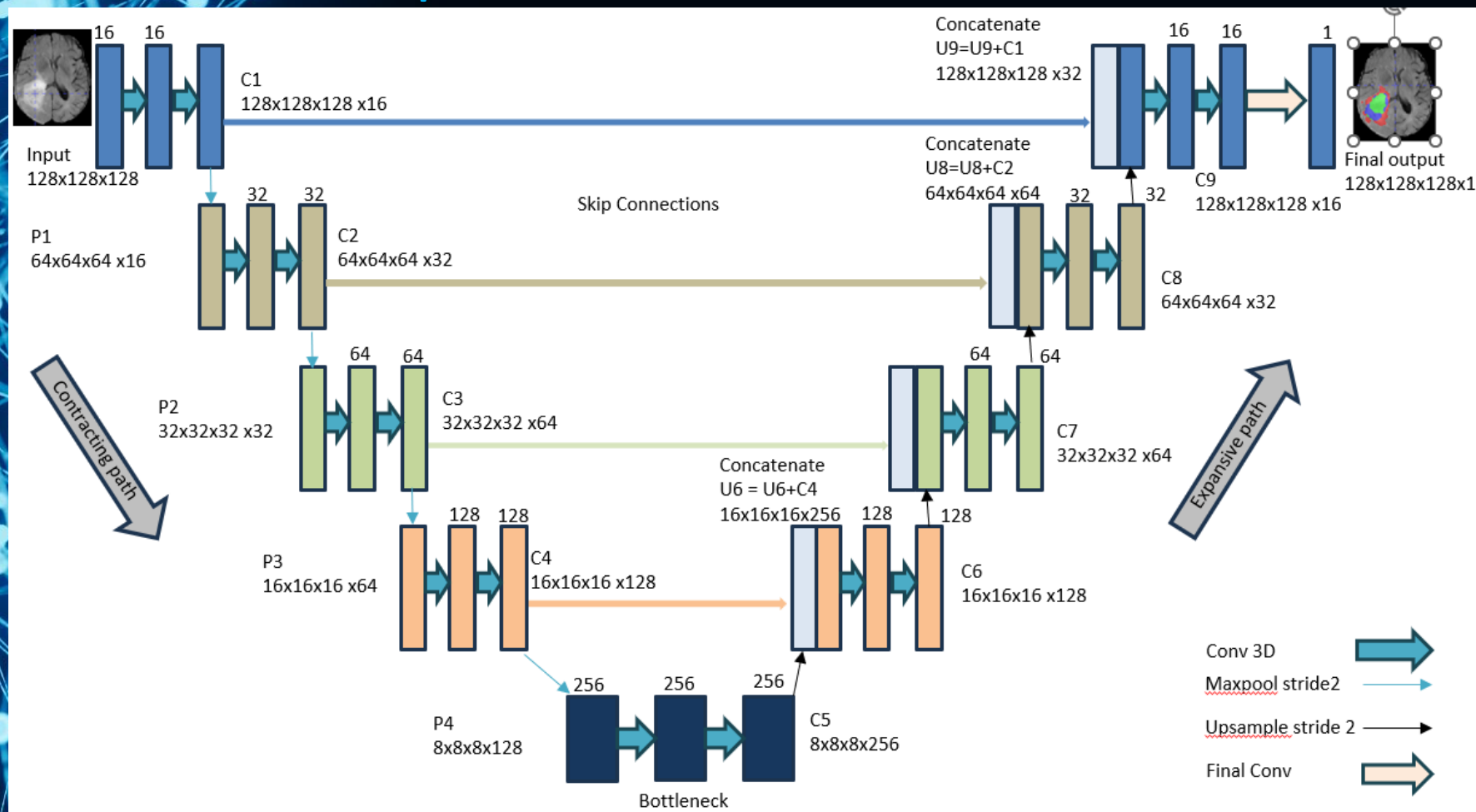
- Catches the tumor but requires more investigation and refinement of the architecture to segment the full region and handle false positives.

The background of the slide features a complex, abstract network of glowing blue nodes and connecting lines, resembling a molecular structure or a data network, set against a dark blue gradient.

Supervised - 3D UNET

- U-Net is a CNN architecture originally designed for biomedical image segmentation, featuring a symmetric “U-shaped” encoder–decoder.
- The encoder (contracting path) repeatedly applies convolutional blocks and down-sampling to capture context and extract deep features.
- The decoder (expansive path) uses transposed convolutions to up-sample feature maps and gradually recover spatial resolution.
- Skip connections link each encoder layer to its corresponding decoder layer, concatenating feature maps to preserve fine-grained localization.
- With its efficient use of data and strong localization ability, U-Net has become a foundational model for medical and semantic segmentation tasks.

Supervised – 3D UNET Architecture



3D UNET with Monai

•Encoder:

- Input $\rightarrow$ 2×ResUnit @16 $\rightarrow$ ↓
- 16 $\rightarrow$ 2×ResUnit @32 $\rightarrow$ ↓
- 32 $\rightarrow$ 2×ResUnit @64 $\rightarrow$ ↓
- 64 $\rightarrow$ 2×ResUnit @128 $\rightarrow$ ↓

•Bottleneck :

- 2×ResUnit @256

•Decoder:

- ↑ 2×ResUnit @128 $\leftarrow$ Skip from step 4
- ↑ 2×ResUnit @64 $\leftarrow$ Skip from step 3
- ↑ 2×ResUnit @32 $\leftarrow$ Skip from step 2
- ↑ 2×ResUnit @16 $\leftarrow$ Skip from step 1

•Final:

- 1×1×1 Conv $\rightarrow$ num_classes (=4) channel output

```
model = UNet(  
    spatial_dims=3,  
    in_channels=1,  
    out_channels=num_classes,  
    channels=(16, 32, 64, 128, 256),  
    strides=(2, 2, 2, 2),  
    num_res_units=2,  
) .to(device)
```

3D UNET with Monai

- **Input:** A 128×128×128 MRI volume.
- **Contracting (“Encoder”) Path**
 - **C1:** two 3×3×3 Conv → ReLU layers producing 16 channels.
 - **P1:** 2×2×2 max-pool reduces to 64³.
 - **C2:** 2xResUnit(Conv+ReLU) → 32 channels; **P2** → 32³.
 - **C3:** 2xResUnit(Conv+ReLU) → 64 channels; **P3** → 16³.
 - **C4:** 2xResUnit(Conv+ReLU) → 128 channels; **P4** → 8³.
 - **Bottleneck (C5):** Conv block → 256 channels at 8³.
- **Expansive (“Decoder”) Path**
 - **U6:** 2× up-sampling from 8³→16³, concatenate with C4 features (skip), then Conv block →128 channels.
 - **U7:** up-sample 16³→32³ + C3 skip → 2xResUnit(DeConv+ReLU) →64 channels.
 - **U8:** up-sample 32³→64³ + C2 skip → 2xResUnit(DeConv+ReLU) →32 channels.
 - **U9:** up-sample 64³→128³ + C1 skip → 2xResUnit(DeConv+ReLU) →16 channels.
- **Final Output**
 - A 1×1×1 Conv projects the 16-channel map to a single segmentation mask (128³×1).
- **Key Points**
 - **Symmetry:** encoder and decoder mirror each other in depth and channel counts.
 - **Skip Connections:** direct concatenation of encoder features restores fine spatial detail.
 - **3D Conv & Pool/Upsample:** all operations are volumetric, preserving the 3D context.

```
model = UNet(  
    spatial_dims=3,  
    in_channels=1,  
    out_channels=num_classes,  
    channels=(16, 32, 64, 128, 256),  
    strides=(2, 2, 2, 2),  
    num_res_units=2,  
)
```


3D UNET Packages & libraries

- **PyTorch:**

- Library for tensor operations and GPU acceleration.
- **Dataset, DataLoader:** abstractions to wrap your data and batch/shuffle it efficiently during training.
- **nn:** PyTorch's module for building neural-network layers and models.
- **optim:** Provides optimizers (e.g. Adam), to update model weights via backpropagation (`nn.CrossEntropyLoss`).

- **SimpleITK (sitk):** Utility for reading, writing, and preprocessing medical-image volumes (e.g. NIfTI).

- **Scikit-learn:**

- **train_test_split:** helper to split file lists or arrays into training and validation sets.

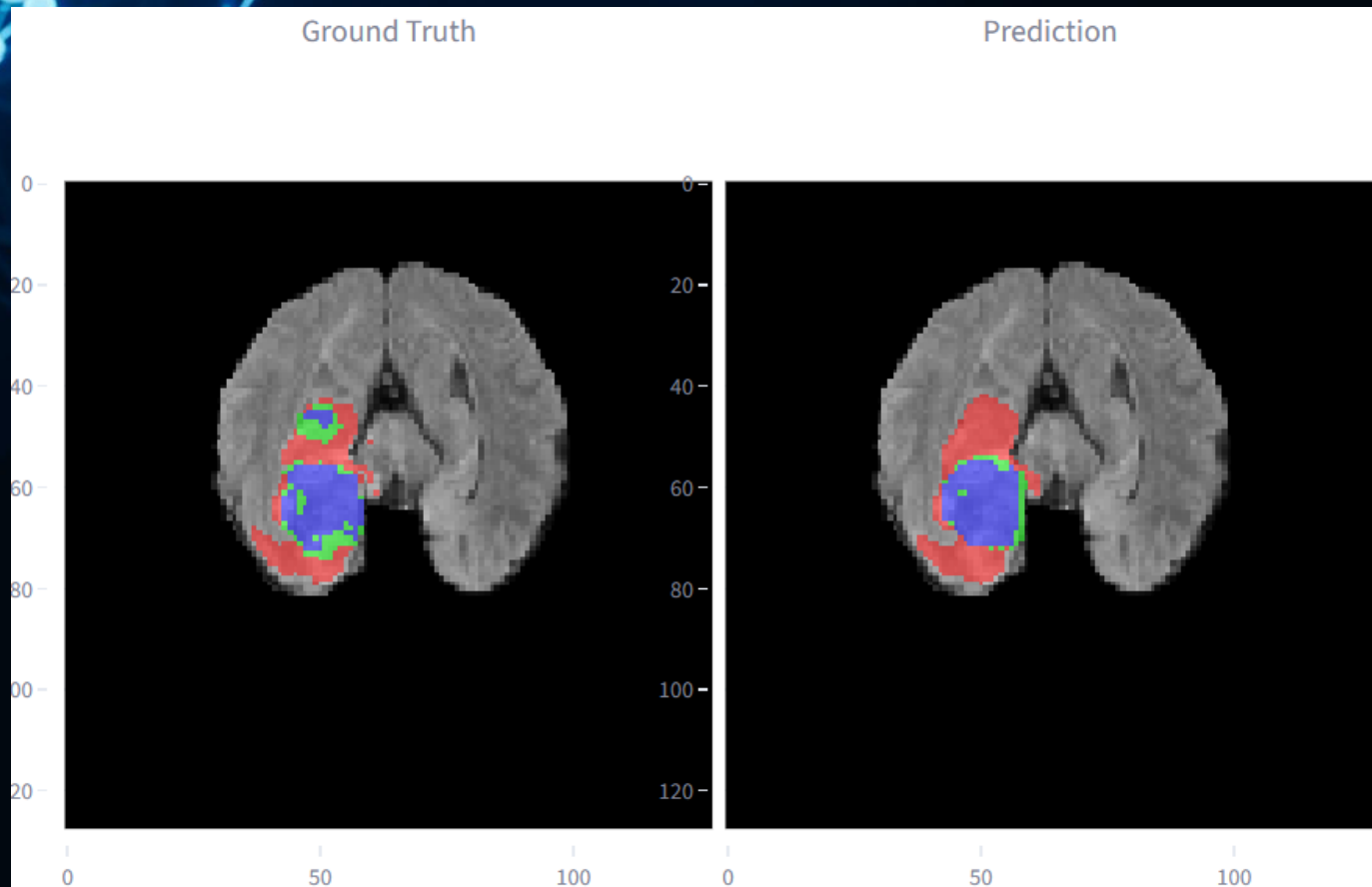
- **UNet (from MONAI):**

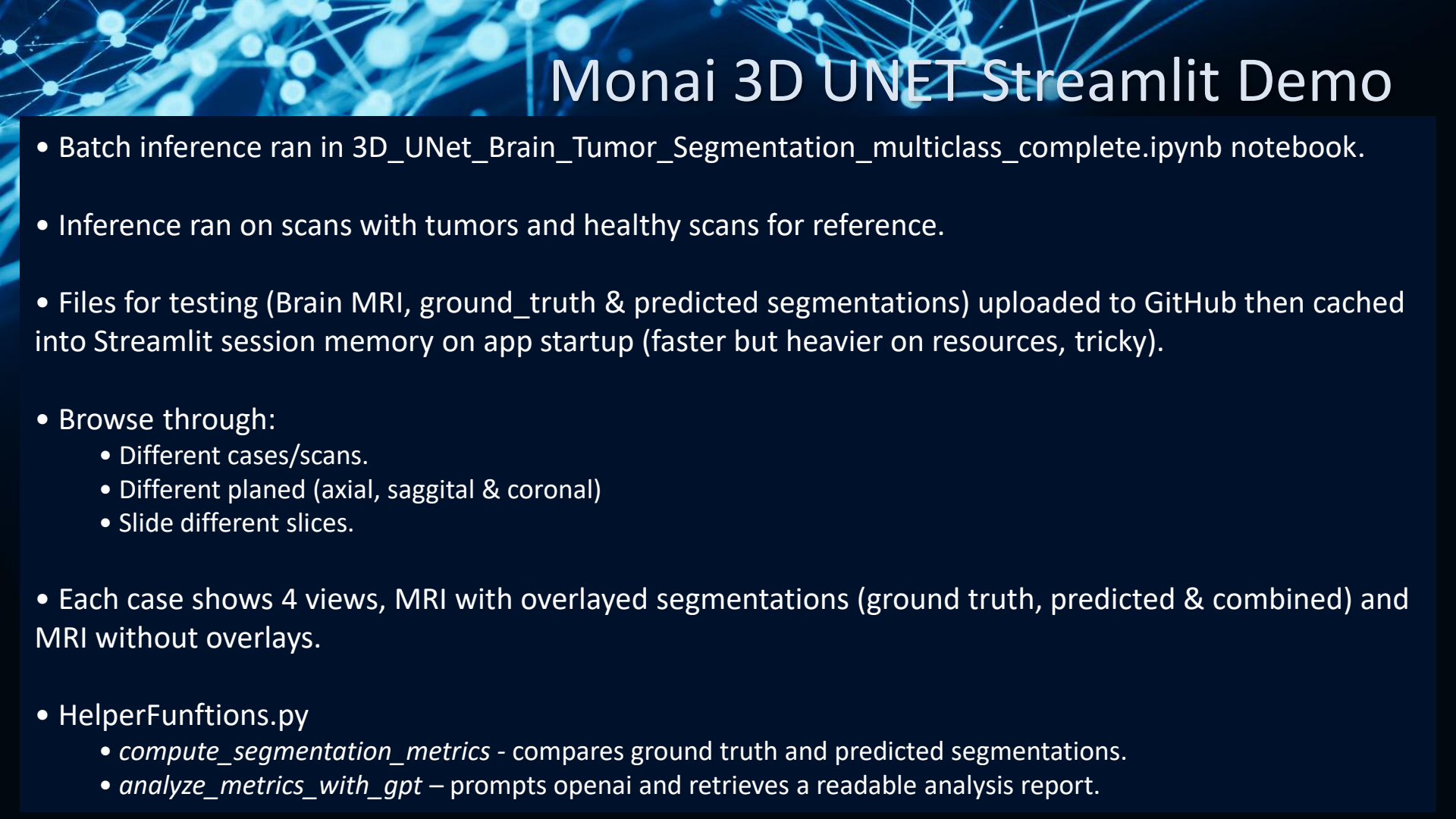
- Off-the-shelf 3D U-Net implementation tailored for medical segmentation.
- **DiceLoss:** A segmentation loss function that maximizes overlap (Dice coefficient) between prediction and ground truth

Monai 3D UNET Batch Inference

- **Setup directories** for tumor or healthy test scans and their corresponding output folders, creating them if needed.
- **Load the trained model** once from **checkpoint_path** and switch to evaluation mode.
- **Loop over all *-T1.nii.gz files** in input_dir, stripping the suffix to build each output filename.
- **Read each volume** with SimpleITK, convert to a NumPy array, and **z-score normalize** it.
- **Convert** the array to a $1 \times 1 \times D \times H \times W$ PyTorch tensor on the chosen device and **run inference** under torch.no_grad().
- **Compute the segmentation** by taking **argmax** over the **softmax** channels.
- **Reconstruct a NIfTI** from the label map (preserving original spacing/origin) and **write** it to output_dir, logging each save.

Monai 3D UNET Batch Inference





Monai 3D UNET Streamlit Demo

- Batch inference ran in 3D_UNet_Brain_Tumor_Segmentation_multiclass_complete.ipynb notebook.
- Inference ran on scans with tumors and healthy scans for reference.
- Files for testing (Brain MRI, ground_truth & predicted segmentations) uploaded to GitHub then cached into Streamlit session memory on app startup (faster but heavier on resources, tricky).
- Browse through:
 - Different cases/scans.
 - Different planed (axial, saggital & coronal)
 - Slide different slices.
- Each case shows 4 views, MRI with overlayed segmentations (ground truth, predicted & combined) and MRI without overlays.
- HelperFunftions.py
 - *compute_segmentation_metrics* - compares ground truth and predicted segmentations.
 - *analyze_metrics_with_gpt* – prompts openai and retrieves a readable analysis report.

Monai 3D UNET Segmentation Metrics

Metric	Short description
$Dice = \frac{2 gtVol \cap predVol }{ gtVol + predVol }$	Measures similarity between ground truth & predicted volumes; 0-1(identical)
$Jaccard = \frac{ gtVol \cap predVol }{ gtVol \cup predVol }$	Ratio of intersection to union, 0 – 1, stricter than Dice.
<i>Precision</i>	High when few false positives.
<i>Recall</i>	High when few false negatives.
<i>F1</i>	FP vs FN trade-off.
$RVD = \frac{ predVol - gtVol }{ gtVol }$	Positive when $ predVol > gtVol $, 0-ideal
<i>Hausdorff Distance</i>	The maximum surface-to-surface error, captures the worst-case boundary mismatch.
<i>HD95 (95th-Percentile Hausdorff)</i>	95th percentile of all point-to-surface distances instead of the maximum, reduces sensitivity to outliers.



3D UNET GBM Auto Segmentation

Next steps:

- Continue investigation on unsupervised strategies & models.
- Continue development & productization with 3D UNET automatic segmentation.



This is just the beginning...

Thank you!

